

DATA ENGINEERING SERIES



# APACHE SPARK

Architecture Explained

Driver

Cluster Mgr

Executors

Full Flow

**Interview Ready in 5 Slides**

Swipe to start learning



# Apache Spark Architecture

2 / 6

## The 3 Core Players — How They Connect

### DRIVER

### The Brain

- Creates SparkSession — entry point for all Spark operations
  - Builds the DAG (execution plan) from your code
- Your .py file runs here — Driver only plans, never processes big data*

requests resources

### CLUSTER MANAGER

### Resource Allocator

- Decides: how many Executors, how much memory
- Supports: **YARN** **Kubernetes** **Standalone**

launches executors on worker nodes

#### Worker Node 1

##### Executor

Task 1

Task 2

Processes data partitions  
Stores in memory/disk

#### Worker Node 2

##### Executor

Task 3

Task 4

Processes data partitions  
Stores in memory/disk

#### Worker Node 3

##### Executor

Task 5

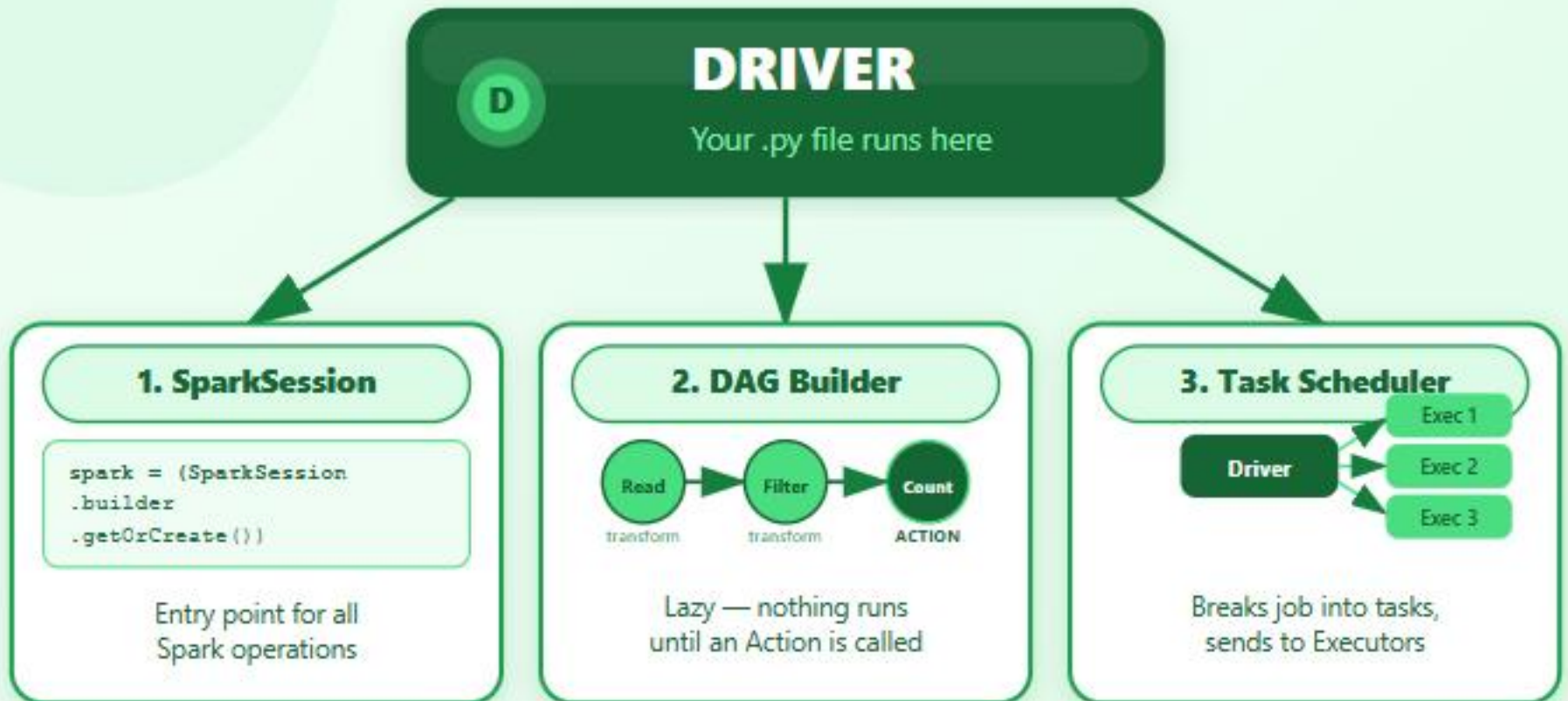
Task 6

Processes data partitions  
Stores in memory/disk

### Key Interview Point

Driver plans — Cluster Manager allocates — Executors execute

Swipe for deeper dives into each component



## How It All Works Together



## Interview Tip

"Driver only plans and coordinates — it never processes big data directly"

## Remember This Code:

```
spark = SparkSession.builder.appName("App").getOrCreate()
```

This one line starts the Driver — the brain of every Spark job



## Driver

"I need resources to run my job"



## CLUSTER MANAGER

Allocates all resources your Spark job needs

*It decides WHAT, HOW MUCH, and WHERE*

How many Executors?

### Executor Count

Exec 1

Exec 2

Exec 3

Based on job size  
and available nodes  
*e.g. --num-executors 10*

How much memory each?

### Memory Allocation

Executor Memory 70%

Overhead 30%

Per-executor memory set  
*e.g. --executor-memory 4g*

Which nodes to use?

### Node Placement



Picks available nodes  
*Cluster-aware scheduling*

## Choose Your Cluster Manager

### YARN

Hadoop Ecosystem

Best for existing Hadoop clusters

*Most common in enterprise*

### Kubernetes

Cloud Native

Best for containerized workloads

*Modern cloud deployments*

### Standalone

Built-in Manager

Comes with Spark — no extras

*Simple dev/test setups*

## Interview Tip

"CM is the middleman — Driver asks, CM decides, Executors are launched"

# Executors — The Real Workers

5 / 6

Each executor runs on a worker node and processes data partitions



## 1 Task = 1 Partition of Data

Each task processes exactly one data partition in parallel

### What Executors Do

- Run tasks assigned by Driver
- Cache data in memory/disk
- Return results to Driver

### Lifecycle

- Start when app starts
- Run for app's lifetime
- Die when app ends

### Interview Tip

- More partitions = more tasks
- = better parallelism
- Partition count matters!

### Remember

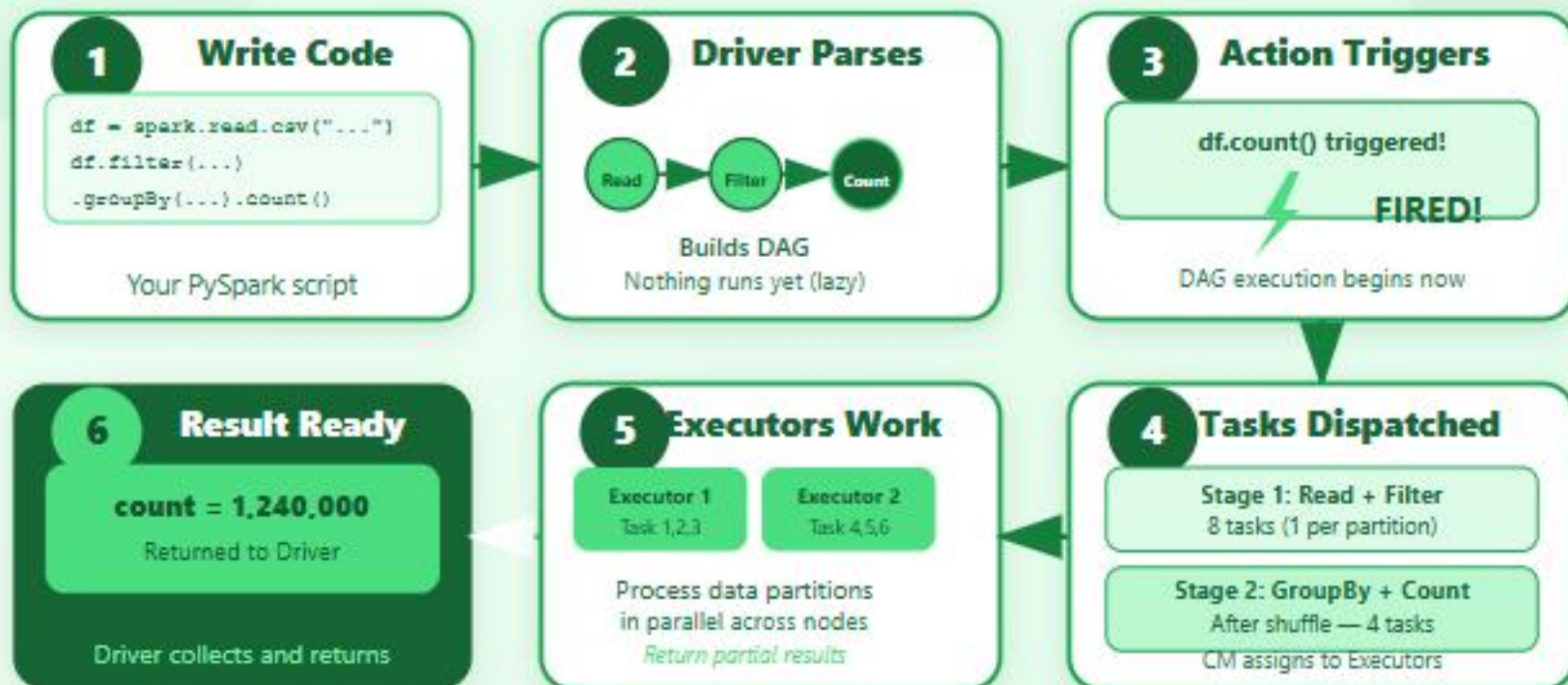
Executors are the **ONLY** component that actually touches your data



# End-to-End Flow — The Full Picture

6 / 6

How your PySpark code becomes results — step by step



## Quick Recap — Who Does What?

|                 |                                                |
|-----------------|------------------------------------------------|
| Driver          | Plans the job, builds DAG, schedules tasks     |
| Cluster Manager | Allocates memory, cores, and executor nodes    |
| Executors       | Process data partitions, store, return results |

## Interview Cheat Sheet

Lazy eval: transformations build DAG, actions trigger execution  
Shuffle (groupBy/join) = data moves across network = slow

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